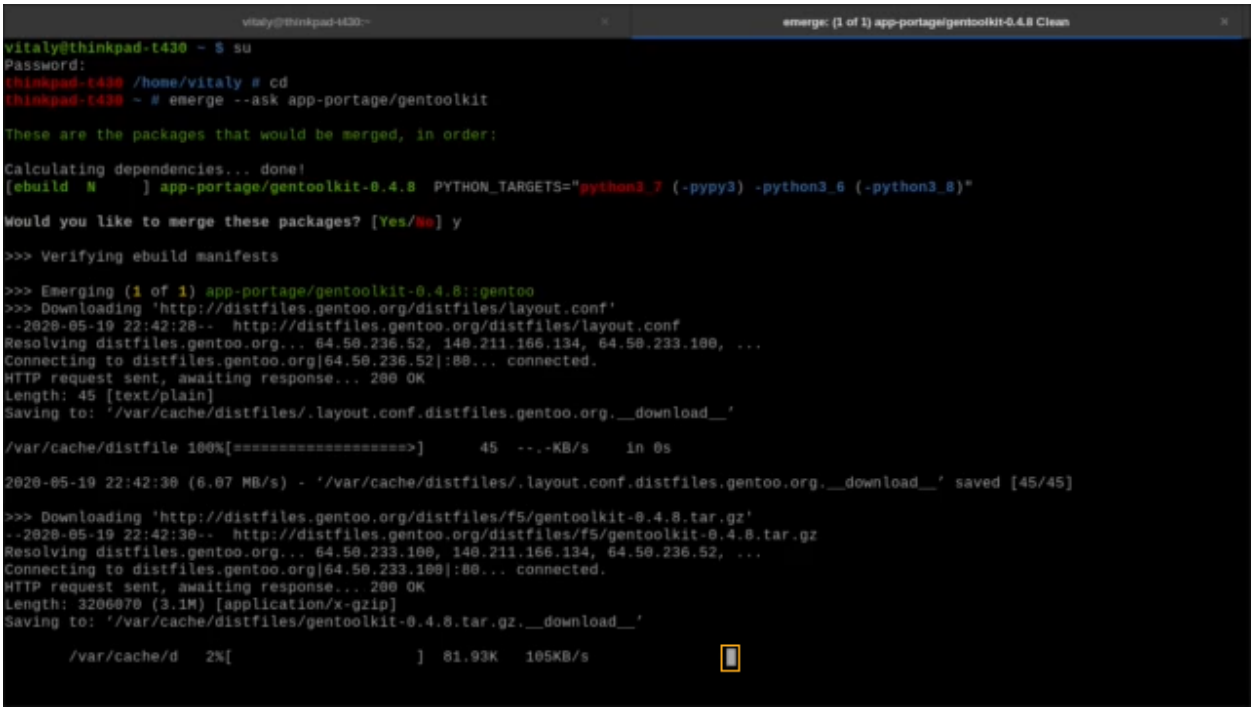
Frame Review: Keep the Lesson

Joseph Ayanda / OpenCV AI Competition 2026 / 23 October / v5 matched boundaries



Actual new Gentoo package-management lesson. Vitaly Zdanevich, via Wikimedia Commons, CC0 as listed. Seconds 62-74 prepared/resized; naturally occurring idle caret highlighted. No defect added or author endorsement.

The concrete problem. A teacher has a flattened screen recording, but no editable project. A caret keeps blinking beside the explanation. Re-recording could disturb the lesson timeline. Frame Review lets the editor hold an actual observed state in one small box and interval, protect the teaching areas and compare the exported file.

Measured now	Scope
Seven original-source HTTP repairs	Six prior regressions plus one new Gentoo source on unchanged frozen code.
90 -> 56 whole-excerpt changes; zero new events	Same 72 seconds in the six prior excerpts; eight-code ROI-mean RGB proxy.
Teaching pixels / timing / prepared audio or absence exact	Independently decoded, all 360 protected frames per repair.
14 verification stages; 3 AWS original browser flows	Jupyter, npm basics and new Gentoo repeat local sources.

Real creator preference, time saved, comprehension, attention, accessibility, safety and adoption have not been measured. An editor can reasonably retain a caret as useful teaching information. The product is a checked experiment requiring review.

The cut points are part of the edit

Observed failure. Fixed four-second holds removed interior activity but could introduce a worse return. Jupyter v4 introduced a selected-mean exit of 0.23028 versus 0.000209 at its old cut. The new cut is elsewhere: frames 19-97, reference 67, rather than 0-119/reference 98. Its return is 0.000934 versus source 0.227989 at that new cut. These are different cut points, not a same-cut improvement claim.

Small causal correction. The stationary detector stays steady-proposal-v2. A separate steady-boundaries-v1 policy tries up to 33 observed reference samples from the originally kept appearance class and trims at most 30 frames per end. It never extends a detected interval, blends or invents pixels. At least 60 held frames and four internal changes must remain. Clip endpoints stay unchanged. It ranks source-compatible cuts by predicted RGB RMS and then useful duration.

Independent encoded gate. Entry, return and loop are measured after actual H.264 export in both selected and four-pixel influence areas. Each RGB mean absolute, RMS and maximum difference must be no worse than source (1e-12 numerical tolerance). The frozen linear selected-mean diagnostic must also be no worse. Whole-excerpt activity must fall with no newly timed eight-code events. Failed gates withhold the edited download; the decision records the measurements.

Exact preservation. Approval binds source, detections, both implementation hashes, box, frames, observed reference and named teaching rectangles. The worker recomputes it. Native outside-box samples, intended reference mapping, all outside-interval frames, all protected decoded RGB samples, timestamps and prepared AAC packet/timing/PCM or audio absence are checked. Original upload and prepared input remain separately recoverable.

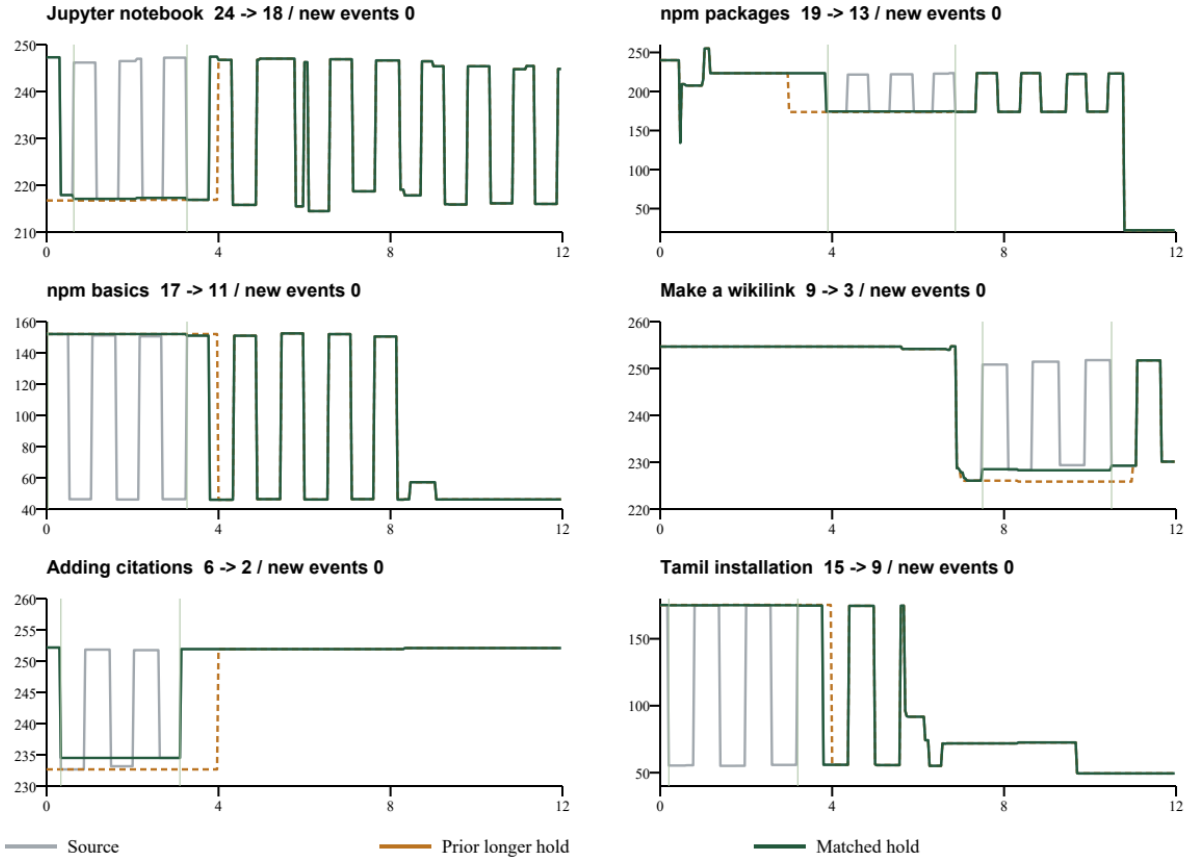
Tradeoff	Retained evidence
New holds 17.6 s versus prior 24 s	Source / fixed blur / prior hold / matched hold: 90 / 91 / 54 / 56 whole-excerpt changes.
Prior hold introduces new events in 5 of 6 cases	Matched hold introduces none. Fixed blur introduces new events in two cases.
26 -> 0 new interval-interior changes	Old 41 -> 0 described different intervals; not interchangeable.
Existing strong cuts may remain	Gentoo entry remains about 0.21815 -> 0.21686. No worsening does not mean smooth or safe.

One actual encoding of the old bad Jupyter hold is now a rejection integration regression. A proposal or successful HTTP request alone is never counted as a qualified repair.

Same-excerpt measured comparison

Same 72 seconds. Fewer changes. No new cut events.

Actual decoded ROI-mean RGB / full 12-second excerpts, including entry and return



New cut points shown in pale green. Each panel has its own labelled RGB range.

Eight-code activity proxy; no measured learning, attention, time saving or safety benefit.

Source	New frames / ref	Source / blur / old / new
Jupyter notebook	19-97 / 67	24 / 24 / 17 / 18
npm packages	117-205 / 116	19 / 20 / 13 / 13
npm basics	1-97 / 0	17 / 18 / 11 / 11
Make a wikilink	225-314 / 251	9 / 8 / 3 / 3
Adding citations	10-92 / 79	6 / 6 / 1 / 2
Tamil installation	6-95 / 24	15 / 15 / 9 / 9

All counts cover the full 12-second prepared excerpt and the same selected box. The blur is one untuned 5x5/sigma-2 native-luma Gaussian inside the new interval, source chroma retained. Actual encoded outputs were decoded. Continuous traces, pixel temporal MAE/MSE, contrast and new-event frames are published. This is not a professional-editor benchmark or human benefit study.

More real lessons, with every outcome

The complete v5 source manifest was frozen before the new-source tests. Four broad sources / 12 quarter-half-three-quarter windows were selected first. After those yielded no useful stationary repair, three targeted command-line tutorials / 9 windows were selected by title, size and listed license before viewing. The follow-up is exploratory, not held-out or representative recall.

Source	All preselected outcomes
Ezhil loops and turtle	3 review windows, no proposal.
GeoGebra matrix	3 original-dimension refusals; declared limit unchanged.
Phoneme table	2 review/no-proposal; 1 >16-region refusal.
WikiSound recording	2 no-proposal; 1 with 2 rapid candidates in instructional Audacity controls/readouts. Both refuse when protected; original/prepared hash recovery passes.
Git Foundations setup	2 no-proposal; 1 with 8 rapid candidates in the instructional webpage. Declined; zero stationary proposals.
Toybox true/false	3 review windows, no proposal.
Gentoo equery	2 no-proposal; 1 stationary caret. Full original HTTP repair qualified, without changing frozen code.

The additional repair. Gentoo original SHA-256 28b6582333d38df21ac89e7511b506bbee9e1d211ad5809a4e729fde5a564eb1. Seconds 62-74; box [370,330,6,10], frames 61-173, reference 86. Whole-excerpt activity 13 -> 9; interior 4 -> 0; zero new events. Commands and changing live progress readout are protected separately and exact in all 360 frames. All boundary/native/timing gates pass. The source is silent; audio absence stays exact. No defect was added.

Two previously negative Info-farmer editor recordings were separately screened over 12 nonoverlapping windows with no proposals. These are offline preparation/detector screens, not HTTP repair qualifications. The convenience cohorts establish neither absence of distractions nor semantic detection accuracy.

Preserved failures. A development Wikilink run hit the unchanged 120 CPU-second budget; redundant decoding was removed, with the final case using about 33.29 CPU seconds. A faster cropped colour transform changed frozen floating-point results and was rejected. The initial full gate hit an unchanged frozen PID-marker test race; one full rerun passed. Cohort orchestration initially hit eight job slots and was split into batches without loosening the app cap.

A complete media path on AWS

Browser original upload -> bounded FFmpeg preparation -> OpenCV 5 localization and observed-state checks -> source-matched cuts -> named teaching protection -> explicit plan approval -> native observed-state export -> independent encoded gates -> owned artifact downloads. Static PNG inspection avoids autoplay; actual downloaded videos remain inspectable.

Substantive OpenCV. Connected components localize temporal support. Native luma measurements test separation, bounded appearance error, dwell and repeated states; chroma stability limits what can be held. OpenCV generates overlays and evaluates encoded RGB errors/histograms in named teaching areas and cut-point boxes. NumPy represents native samples. FFmpeg performs bounded preparation, exact lossless encoding and prepared-audio transport.

Live release fact	Verified value
Hosting	ECS Express / Fargate / managed HTTPS / CloudWatch
Image digest	sha256:b47393cbfdb45d0be950062a89d26b10e114d0b36acaedfb3a6d52368706c543
Task	default-lesson-rescue:10 / one vCPU / 2 GiB / no application AWS role
Runtime	a900388b23d5cdab6f2ad590a669fed2d70e50eee32ef20891b33a7249ca5ac1 / 106 exact files
Original browser checks	Jupyter, npm basics, Gentoo: real original uploads, explicit review/approval, 7 artifact and 12 crop downloads each; report/recipe/source hashes match.
User-facing checks	Reload recovers completed jobs; 375/890 layouts have no horizontal overflow; downloaded videos play to 12 s at 640x360, no media error. Playback muted; no listening claim.
Ownership	Known-job requests without the owner cookie return 404; one worker / 8 slots / 24-hour ephemeral task-local jobs.

AWS browser runs repeat the same locally qualified sources. Downloaded edited bytes match independent local qualifications. The single-task rollout is complete; cloud evidence is distinct from local tests. Ephemeral jobs can disappear on task replacement. The profile remains 1-12 seconds, 640x360/30 fps, up to 64 MiB original and declared supported SDR tracks; preparation changes picture/audio and can reduce legibility. No full-lesson splice-back or semantic tracking is implemented.

The user-facing innovation is a small, source-observed editing decision with reviewable receipts and refusal. The impact hypothesis is avoiding a re-record of a short explanation; it needs consenting creator preference and timed-task studies. The official rubric weights execution, innovation, impact, UX, documentation and cloud reproducibility separately. No judge score, competition win or special-award qualification is claimed.

Reproduce, inspect, attribute

Unzip the submitted source package and enter Frame-Review. Build: `docker build -t frame-review .` Run the full gate in a new container: `docker run --name frame-review-check --network none frame-review python studio/run.py --verify /tmp/qualification`. Copy receipts with docker cp, then serve with `docker run --rm --network host frame-review python studio/run.py --serve --port 8877`. The Linux process/resource controls are part of media qualification; Windows source/UI checks alone are insufficient. Use fresh output directories.

All fourteen gate stages pass. Backend 137 unit tests; studio 84 tests, including one platform skip. Verification JSON is authoritative; the CLI last sentence retains its older word “Thirteen.” The new boundary rejection stage actually encodes the former bad Jupyter hold. The frozen opencv-ready package and older failure evidence stay unchanged.

For original-upload reproduction, use evidence/natural-v5/sources.json URLs and SHA-256, the six source settings in docs/natural-v5/natural-v4-cases.json, and natural-v5-gentoo-case.json. The historical configuration filename does not select old code. The current qualifier recomputes the frozen v5 proposal and independently checks decoded samples, boundaries, audio, source recovery and all 12 crop downloads. Comparison and both source-screen harnesses are included. Fresh named native evidence directories are required.

Source / creator	Listed rights and adaptation
Jupyter / Alex The Analyst; npm packages / Ganesh H; npm basics / Kevin Powell	Commons CC BY 3.0. Imported YouTube license reviews listed not yet reviewed.
Wikilink / Psypherium	Commons CC BY 3.0.
Adding citations / James Gaunt + Wikimedia Foundation material	CC BY 4.0 plus embedded CC BY-SA 3.0; adapted asset CC BY-SA 4.0. Both credits preserved.
Tamil installation / Info-farmer; Gentoo / Vitaly Zdanevich	Commons CC0.
All new-screen sources	Exact authors, license URLs, original hashes and selection plans in sources.json. Metadata is attribution, not independent clearance or endorsement.

Prepared excerpts and comparison videos are adaptations: extraction, resizing, H.264 conversion, one-time optional AAC preparation and the declared observed-state hold. Originals are separately attributed and not bundled. No installed dependency binaries, credentials, owner cookies, live job directories or handoff documents are in the source package. The existing 2:46 neural-narrated video covers an earlier workflow; Joseph’s real-voice recording is deferred. Current v5 evidence is in this report, the live app, exact source archive and receipts.